

Research Profile

I design and build **multimodal sensemaking systems for health and well-being**, pursuing **human-AI co-sensemaking**: a bidirectional process in which AI helps people interrogate and interpret their health data while they, in turn, shape the system's understanding of their context and goals. My goal is to close the **action gap**, turning what sensors capture and models infer into safe, concrete next steps for clinicians and individuals. By integrating wearable and ubiquitous sensing, machine learning and generative AI, and interactive visual analytics, and working through sustained clinical collaborations, I build and evaluate systems for behavioral health, clinical decision support, and everyday well-being. My research lies at the intersection of **Ubiquitous Computing (UbiComp)**, **Human-Computer Interaction (HCI)**, and **Health Informatics**.

Keywords: Human-AI Co-Sensemaking, Multimodal Health Sensing, Generative AI Systems, Clinical Decision Support, Interactive Visual Analytics, Digital Well-being

Employment

Postdoctoral Research Scientist	Department of Biomedical Informatics, Columbia University Advisor: <u>Prof. Xuhai "Orson" Xu</u>	2026–Present
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Education

Ph.D.	The University of Hong Kong , Department of Electrical and Computer Engineering Advisor: Prof. Edith C.H. Ngai Thesis: <i>"Empowering Pervasive Healthcare: Mobile Analytics Systems Leveraging Multimodal Data"</i> (Excellent PhD Thesis)	2021–2025
M.E.	Xiamen University , Department of Computer Science and Technology Advisor: Prof. Longbiao Chen Thesis: <i>"Traffic Violation Hotspot Identification, Diagnosing, and Prediction Based on Spatial Crowdsensing Big Data"</i> (Outstanding Graduate Thesis)	2018–2021
B.E.	Xiamen University Major: Computer Science and Technology, School of Informatics Thesis: <i>"Identification and Visualization of Driving Violations Based on Large-Scale Vehicle Trajectory Data"</i> (Outstanding Undergraduate Thesis) Minor: Mathematical Finance, The Wang Yanan Institute for Studies in Economics	2014–2018

Publications (* indicates equal contribution, † indicates corresponding author)

Full Peer-Reviewed Conference & Journal Papers

2026

- P35. [ACM HEALTH'26] **Zhihan Jiang***, Running Zhao*, Lin Lin, Yue Yu, Handi Chen, Xincheng Zhang, Xuhai Xu, Yifang Wang, Xiaojuan Ma, Edith C.H. Ngai. DietGlance: Dietary Monitoring and Personalized Analysis at a Glance with Knowledge-Empowered AI Assistant. *ACM Transactions on Computing for Healthcare*, 2026.
- P34. [CHI'26] **Zhihan Jiang**, Qianhui Chen, Chu Zhang, Yanheng Li, Ray LC. Hear You in Silence: Designing for Active Listening in Human Interaction with Conversational Agents Using Context-Aware Pacing. *ACM CHI Conference on Human Factors in Computing Systems*, 2026. ( **Honorable Mention Award Award, 5%**)
- P33. [CHI'26] Mengyuan Wu, **Zhihan Jiang***, Yuang Fan*, Richard Feng, Sahiti Dharmavaram, Mathew Polowitz, Shawn Fallon, Bashima Islam, Lizbeth Benson, Irene Tung, David Creswell, Xuhai Xu. MindfulAgents: Personalizing Mindfulness Meditation via an Expert-Aligned Multi-Agent System. *ACM CHI Conference on Human Factors in Computing Systems*, 2026. ( **Honorable Mention Award Award, 5%**)

- P32. [TKDD'26] Jing Deng, Handi Chen*, **Zhihan Jiang***, Raymond Chi-Wing Wong, Edith C.H. Ngai. A Survey on Causality with Federated Learning: Challenges, Techniques, and Applications. *Transactions on Knowledge Discovery from Data*, 2026.
- P31. [ToN'26] Handi Chen, Jing Deng, Xiuzhe Wu, **Zhihan Jiang**, Xincheng Zhang, Xianhao Chen, Edith C.H. Ngai, Jiangchuan Liu. LiFeChain: Lightweight Blockchain for Secure and Efficient Federated Lifelong Learning in IoT. *IEEE Transactions on Networking*, 2026.

2025

- P30. [UIST'25] Running Zhao*, **Zhihan Jiang***, Xincheng Zhang, Chirui Chang, Handi Chen, Weipeng Deng, Luyao Jin, Xiaojuan Qi, Xun Qian, Edith C.H. Ngai. NoteIt: A System Converting Instructional Videos to Interactable Notes Through Multimodal Video Understanding. *The ACM Symposium on User Interface Software and Technology*, 2025.
- P29. [AFM'25] Ran An, **Zhihan Jiang**, Qiyu Cao, Shuang Zhang, Edith C. H. Ngai, Tianshuo Zhao. Chemically Programmable Fano Resonances via Colloidal Nanocrystal-Ligand Chemistry for Ultra-Sensitive Ion Detection. *Advanced Functional Materials*, 2025.
- P28. [TNSE'25] Zhiyi Zhong, Lin Lin, **Zhihan Jiang**, Xin Yuan, Edith C.H. Ngai, James Lam, Ka-Wai Kwok. Connectivity Determination Algorithm for Complex Directed Networks. *IEEE Transactions on Network Science and Engineering*, 2025.
- P27. [INFOCOM'25] Xincheng Zhang, Running Zhao, **Zhihan Jiang**, Handi Chen, Yulong Ding, Edith C.H. Ngai, Shuanghua Yang. Continual Learning with Strategic Selection and Forgetting for Network Intrusion Detection. *The 2025 IEEE International Conference on Computer Communications*, 2025.
- P26. [TMC'25] Running Zhao, Jiangtao Yu, Tingle Li, **Zhihan Jiang**, Chenwei Zhang, Cunshu Wu, Hang Zhao, Edith C.H. Ngai. SPACE: Speaker Adaptation for Acoustic Eavesdropping using mmWave Radio Signals. *IEEE Transactions on Mobile Computing*, 2025.
- P25. [IEEE Network'25] Handi Chen, Weipeng Deng, Shuo Yang, Jinfeng Xu, **Zhihan Jiang**, Edith C.H. Ngai, Jiangchuan Liu, Xue Liu. Towards Edge General Intelligence via Large Language Models: Opportunities and Challenges. *IEEE Network*, 2025.

2024

- P24. [JAACAPOpen'24] **Zhihan Jiang***, Adrienne Y.L. Chan*, Dawn Lum, Kirstie HTY Wong, Janice CN Leung, Patrick Ip, David Coghill, Rosa S Wong, Edith C.H. Ngai, Ian C.K. Wong. Wearable Signals for Diagnosing Attention-Deficit/Hyperactivity Disorder in Adolescents: A Feasibility Study. *Journal of American Academy of Child & Adolescent Psychiatry Open*, 2024.
- P23. [JFranklin'24] Xincheng Zhang, **Zhihan Jiang**, Yulong Ding, Edith C.H. Ngai, Shuang-Hua Yang. Anomaly Detection using Isomorphic Analysis for False Data Injection Attacks in Industrial Control Systems. *Journal of the Franklin Institute*, 2024.
- P22. [INFOCOM'24] Xincheng Zhang, Running Zhao, **Zhihan Jiang**, Zhicong Sun, Yulong Ding, Edith C.H. Ngai and Shuanghua Yang. AOC-IDS: Autonomous Online Framework with Contrastive Learning for Intrusion Detection. *The 2024 IEEE International Conference on Computer Communications*, 2024.
- P21. [TMC'24] Handi Chen, Rui Zhou, Yun-Hin Chan, **Zhihan Jiang**, Xianhao Chen, Edith C.H. Ngai. LiteChain: A Lightweight Blockchain for Verifiable and Scalable Federated Learning in Massive Edge Networks. *IEEE Transactions on Mobile Computing*, 2024.
- P20. [ICLR'24] Yun-Hin Chan, Rui Zhou, Running Zhao, **Zhihan Jiang**, Edith C.H. Ngai. Internal Cross-layer Gradients for Extending Homogeneity to Heterogeneity in Federated Learning. *The 12th International Conference on Learning Representations*, 2024.
- P19. [UbiComp/IMWUT'24] Yufei Wang, Wenting Zeng, Changzhen Liu, Zhuohan Ye, Jiawei Sun, Junxiang Ji, **Zhihan Jiang**, Xianyi Yan, Yongyi Wu, Yigao Wang, Dingqi Yang, Leye Wang, Daqing Zhang, Cheng Wang, Longbiao Chen. CrowdBot: An Open-Environment Robot Management System for On-Campus Services. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2024.

2023

- P18. [IoTJ'23] **Zhihan Jiang**, Vera van Zoest, Weipeng Deng, Edith C.H. Ngai, Jiangchuan Liu. Leveraging Machine Learning for Disease Diagnoses based on Wearable Devices: A Survey. *IEEE Internet of Things Journal*, 2023.

- P17. [TVCG/VIS'23] **Zhihan Jiang**, Handi Chen, Rui Zhou, Jing Deng, Xincheng Zhang, Running Zhao, Cong Xie, Yifang Wang, Edith C.H. Ngai. HealthPrism: A Visual Analytics System for Exploring Children's Physical and Mental Health Profiles with Multimodal Data. *IEEE Transactions on Visualization and Computer Graphics*, IEEE VIS 2023.
- P16. [UbiComp/IMWUT'23] **Zhihan Jiang**, Lin Lin, Xincheng Zhang, Jianduo Luan, Running Zhao, Longbiao Chen, James Lam, Ka Man Yip, Hung Kwan So, Wilfred HS Wong, Patrick Ip, Edith C.H. Ngai. A Data-Driven Context-Aware Health Inference System for Children during School Closures. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023.
- P15. [TCNS'23] Lin Lin, **Zhihan Jiang**, Hong Lin, Edith C.H. Ngai, James Lam. On Quotients of Stochastic Networks over Finite Fields. *IEEE Transactions on Control of Network Systems*, 2023.
- P14. [UbiComp/IMWUT'23] Tieqi Shou, Zhuohan Ye, Yayao Hong, Zhiyuan Wang, Hang Zhu, **Zhihan Jiang**, Dingqi Yang, Binbin Zhou, Cheng Wang, Longbiao Chen. CrowdQ: Predicting the Queue State of Hospital Emergency Department Using Crowdsensing Mobility Data-Driven Models. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023.

2022

- P13. [TITS'22] **Zhihan Jiang**, Xin He, Chenhui Lu, Binbin Zhou, Xiaoliang Fan, Cheng Wang, Xiaojuan Ma, Edith C.H. Ngai, Longbiao Chen. Understanding Drivers' Visual and Comprehension Loads in Traffic Violation Hotspots Leveraging Crowd-Based Driving Simulation. *IEEE Transactions on Intelligent Transportation Systems*, 2022.
- P12. [Healthcare'22] **Zhihan Jiang***, Ka-Man Yip*, Xincheng Zhang, Jing Deng, Wilfred Wong, Hung-Kwan So, Edith C. H. Ngai. Identifying the High-Risk Population for COVID-19 Transmission in Hong Kong Leveraging Explainable Machine Learning. *Healthcare*, 2022.
- P11. [CHI'22] Chuhan Shi, **Zhihan Jiang**, Xiaojuan Ma, Qiong Luo. A Personalized Visual Aid for Selections of Appearance Building Products with Long-term Effects. *Proc. of the ACM CHI Conference on Human Factors in Computing Systems*, 2022.
- P10. [UIC'22] Jiannan Gao, Yigao Wang, **Zhihan Jiang**, Hang Zhu, Qiyue Zhong, Xiaoliang Fan, Longbiao Chen, Cheng Wang. iTA: Inferring Traffic Accident Hotspots with Vehicle Trajectories and Road Environment Data. *The 19th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2022.
- P9. [TMC'22] Hang Zhu, Tieqi Shou, Ruiying Guo, **Zhihan Jiang**, Zeyu Wang, Zhiyuan Wang, Zhiyong Yu, Weijie Zhang, Cheng Wang, Longbiao Chen. RedPacketBike: A Graph-Based Demand Modeling and Crowd-Driven Station Rebalancing Framework for Bike Sharing Systems. *IEEE Transactions on Mobile Computing*, 2022.

2021

- P8. [TMC'21] **Zhihan Jiang**, Hang Zhu, Binbin Zhou, Chenhui Lu, Mingfei Sun, Xiaojuan Ma, Xiaoliang Fan, Cheng Wang, Longbiao Chen. CrowdPatrol: A Mobile Crowdsensing Framework for Traffic Violation Hotspot Patrolling. *IEEE Transactions on Mobile Computing*, 2021.
- P7. [JNCA'21] Longbiao Chen, **Zhihan Jiang**, Dingqi Yang, Thi-Mai-Trang Nguyen, Cheng Wang. Fog Radio Access Network Optimization for 5G Leveraging User Mobility and Traffic Data. *Journal of Network and Computer Applications*, 2021.
- P6. [UbiComp/IMWUT'21] Longbiao Chen, Chenhui Lu, Fangxu Yuan, **Zhihan Jiang**, Leye Wang, Daqing Zhang, Ruixiang Luo, Xiaoliang Fan, Cheng Wang. UVLens: Urban Village Boundary Identification and Population Estimation Leveraging Open Government Data. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2021.
- P5. [UIC'21] Xin He, Hong Hong, Yanwen Liu, Xiaojuan Ma, **Zhihan Jiang**, Longbiao Chen, Ming Cheng, Cheng Wang, Yongchuan Li. CovidPass: A Contactless Check-in System for Keeping Social Distance in Public Health Crisis. *The 18th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2021.

Previous

- P4. [ISJ'20] **Zhihan Jiang**, Longbiao Chen, Binbin Zhou, Jingchun Huang, Tianqi Xie, Xiaoliang Fan, Cheng Wang. iTV: Inferring Traffic Violation-Prone Locations with Vehicle Trajectories and Road Environment Data. *IEEE Systems Journal*, 2020.
- P3. [FCS'20] **Zhihan Jiang**, Yan Liu, Xiaoliang Fan, Cheng Wang, Jonathan Li, Longbiao Chen. Understanding Urban Structures and Crowd Dynamics Leveraging Large-Scale Vehicle Mobility Data. *Frontiers of Computer Science*, 2020.

- P2. [UIC'19] Ruiying Guo, **Zhihan Jiang**, Jingchun Huang, Jianrong Tao, Cheng Wang, Jonathan Li, Longbiao Chen. BikeNet: Accurate Bike Demand Prediction Using Graph Neural Networks for Station Rebalancing. *The 16th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2019.
- P1. [EWSN'19] Longbiao Chen, **Zhihan Jiang**, Jiangtao Wang, Yasha Wang. Data-Driven Bike Sharing System Optimization: State of the Art and Future Opportunities. *The 2019 International Conference on Embedded Wireless Systems and Networks*, 2019.

Peer-Reviewed Workshop, Poster & Short Papers

- W5. [Education Forum] **Zhihan Jiang**, Xuhai “Orson” Xu. From AI-as-Shortcut to AI-as-Partner: Student-AI Collaborative Inquiry for Health-Agent Education. *UbiComp/ISWC 2026 Education Forum*. (to appear)
- W4. [Tutorial] **Zhihan Jiang**, Will Ke Wang, Blue (Georgianna) Lin, Brenna Li, Xuhai “Orson” Xu. From Multimodal Sensing Data to Actionable Health Insights: A Hands-on Guide to Prototyping Your Personal LLM Health Agent. *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, 2026.
- W3. [CHI Workshop'26] **Zhihan Jiang**, Mengyuan “Millie” Wu, Ruishi Zou, Shiyu Xu, Emma Macmanus, Steven Liao, Ping Zhang, Dakuo Wang, James L. David, Nabila El-Bassel, Lena Mamykina, Frances R. Levin, Ryan Sultan, Xuhai Xu. PAWS: Empowering Everyday Cannabis Use Disorder Support through a Personalized AI Digital Pet on Smartwatches. *Everyday Wearable for Personalized Health and Well-Being Workshop @ ACM CHI Conference on Human Factors in Computing Systems*, 2026. (🏆 **Best Poster Award**)
- W2. [ICASSP Workshop'23] **Zhihan Jiang**, Cong Xie, Edith C.H. Ngai. A Health Profiling Framework for Children Leveraging Multimodal Learning Based on Ambient Sensor Signals. *Ambient AI Workshop @ IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2023.
- W1. [HHME'18] Yiyang Lai, **Zhihan Jiang**, Xiaoliang Fan, Cheng Wang, Jun Li, Longbiao Chen. Post-Disaster Traffic Visualization and Analysis based on Large-Scale Vehicle Trajectory Data. *The 14th Joint Conference on Harmonious Human-Machine Environment*, 2018.
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Patents

- T3. Longbiao Chen, **Zhihan Jiang**, Xiaoliang Fan, Cheng Wang, Hong Hong. A Method, System, and Readable Storage Medium for Patrol Route Planning Based on Traffic Violation Hotspot Prediction. China Patent CN202010693026.4.
- T2. Longbiao Chen, **Zhihan Jiang**, Cheng Wang. A Method and System for Traffic Violation Identification Based on Traffic Data and Street View Data. China Patent CN202010031165.0.
- T1. Longbiao Chen, Ruiying Guo, **Zhihan Jiang**, Zhiyuan Wang, Cheng Wang. A Method for Bike Demand Prediction and Scheduling Based on Deep Learning and Crowd Sensing. China Patent CN202010599191.3.
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Selected Scholarships & Honors

Best Poster Award , Everyday Wearable for Personalized Health and Well-Being Workshop @ CHI 2026	2026
Honorable Mention Award (×2), CHI 2026	2026
OPA Academic Conference Travel Grant Award , Columbia University	2026
Google Cloud Research Credit Award	2025
Excellent PhD Thesis , The University of Hong Kong	2025
HKU Foundation First Year Excellent PhD Award (one of no more than 15 recipients university-wide)	2024
IEEE VIS Inclusivity and Diversity Scholarship , IEEE VIS	2023
Conference Grant for RPg Student (twice), The University of Hong Kong	2023
Outstanding Graduate Thesis , Xiamen University	2021
Outstanding Graduates , Xiamen University	2021
National Scholarship (highest honor awarded by the Ministry of Education of China, top 1%)	2020
Outstanding Undergraduate Thesis , Xiamen University	2018
Academic Innovation Scholarship , Xiamen University	2017
Academic Excellence Scholarship (four times), Xiamen University	2015–2018

Academic Service

Editor, Chair, Committee

- **Lead Organizer**, Tutorial on Prototyping Personal LLM Health Agents @ ACM UbiComp/ISWC 2026
- **Technical Program Committee** @ EIoT Workshop, UbiComp/ISWC 2026
- **Social Chair / Technical Program Committee** @ WellComp Workshop, UbiComp/ISWC 2026
- **Session Chair** @ CHI 2026
- **Program Committee** @ UbiComp/ISWC 2026 Poster Track

Journal Reviewer

- **[JMIR]** Journal of Medical Internet Research
- **[HEALTH]** ACM Transactions on Computing for Healthcare
- **[TCE]** IEEE Transactions on Consumer Electronics
- **[TAFFC]** IEEE Transactions on Affective Computing
- **[TVCG]** IEEE Transactions on Visualization and Computer Graphics
- **[TITS]** IEEE Transactions on Intelligent Transportation Systems
- **[TII]** IEEE Transactions on Industrial Informatics
- **[TCSS]** IEEE Transactions on Computational Social Systems
- **[TMC]** IEEE Transactions on Mobile Computing
- **[IMWUT]** Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- **[JNCA]** Journal of Network and Computer Applications
- **[IoTJ]** IEEE Internet of Things Journal

Conference Reviewer

- **[VIS]** IEEE Visualization and Visual Analytics Conference (*reviews recognized as highly useful*) 2026
- **[UIST]** The ACM Symposium on User Interface Software and Technology 2026
- **[IUI]** ACM Conference on Intelligent User Interfaces 2026
- **[PacificVis]** IEEE Pacific Visualization Conference (*reviews recognized as highly useful*) 2026
- **[HAI]** International Conference on Human-Agent Interaction 2026
- **[IJCNN]** International Joint Conference on Neural Networks 2025
- **[CHI]** ACM Conference on Human Factors in Computing Systems 2025–2026
- **[TEI]** ACM Int. Conf. on Tangible, Embedded, and Embodied Interaction 2025
- **[INFOCOM]** IEEE International Conference on Computer Communications 2023–2025
- **[SenSys]** ACM Conference on Embedded Networked Sensor Systems 2022–2025
- **[ICASSP]** IEEE Int. Conf. on Acoustics, Speech, and Signal Processing 2024–2026
- **[ICDCS]** IEEE International Conference on Distributed Computing Systems 2022–2023
- **[IWQoS]** IEEE/ACM International Workshop on Quality of Service 2023
- **[UIC]** IEEE Int. Conf. on Ubiquitous Intelligence and Computing 2020–2022
- **[ChineseCSCW]** Chinese Conference on CSCW and Social Computing 2020
- **[ICC]** IEEE International Conference on Communications 2020

Teaching Experience

Teaching Roles and Responsibilities

- Leading tutorials, designing and grading assignments, and holding office hours.
- Certificate in Teaching and Learning in Higher Education awarded by The University of Hong Kong.

Conference Tutorial

From Multimodal Sensing Data to Actionable Health Insights: A Hands-on Guide to Prototyping UbiComp/ISWC, Oct 2026
Your Personal LLM Health Agent (**Lecturer**)

Guest Lecturer, Columbia University

BINF 4070 / COMS 4995: *The Future of Personal Health Assistant: Multimodal Health Data Management Agents* (graduate course); lecture on LLM agents, tool use, and function calling Fall 2026

Teaching Assistant, The University of Hong Kong

COMP3234 / ELEC3443: Computer and Communication Networks Undergraduate core course, ~70–100 students per year Fall 2021–2024

Teaching Assistant, Xiamen University

Software Architecture and Design Patterns Spring 2019

Invited Talks & Seminars

- Research Summit for Egocentric Intelligence with Project Aria, Meta Reality Labs Research Redmond, WA, USA, Jul 2026
- Southern University of Science and Technology Shenzhen, China, Jun 2026
- Northeastern University Human-Centered AI Lab (NEU HAI) Boston, USA, May 2026
- Dept. of Computer and Computing Science, Hangzhou City University Hangzhou, China, Oct 2025
- Tam Wing Fan Innovation Wing Two, The University of Hong Kong Hong Kong SAR, Jun 2024
- Dept. of Electrical and Computer Engineering, The University of Hong Kong Hong Kong SAR, Oct 2023
- MARS Group, Xiamen University Xiamen, China, Aug 2023

Selected Media Coverage

- **Featured Interview: “Her Power in Science: Ubiquitous Computing Empowers Healthcare”** Mar 2024
Hong Kong Wen Wei Po (Mainstream Press) & The University of Hong Kong (Official Multi-channel Feature)
Selected as a featured doctoral candidate showcasing the societal value of pervasive healthcare research.
- **Research Highlight: AI-Powered Note-taking Tool “NoteIt”** Sep 2025
The University of Hong Kong (Official X/Twitter Feature)
Featured coverage of the team’s innovative multimodal video understanding system.
- **Research Highlight: AI System for Child Health Assessment** Dec 2023
The University of Hong Kong (Official X/Twitter Feature)
Highlighted contributions to designing health inference systems during COVID-19 school closures.